

LEG MUSCLES

PLAN

- I. INTRODUCTION**
- II. DESCRIPTIVE ANATOMY:**
 - A. ANTERIOR COMPARTMENT MUSCLES**
 - B. LATERAL COMPARTMENT MUSCLES**
 - C. POSTERIOR COMPARTMENT MUSCLES**

Objectives

- To know the origin and insertion of the leg muscles.
- To be able to classify these muscles according to their actions.
- To know the innervation of these muscles.

I. INTRODUCTION:

The leg is divided into three muscle compartments: anterior, lateral, and posterior, by two anterior and posterior intermuscular septa.

- Muscles of the anterior compartment: extensors of the toes and flexors of the foot.
- Muscles of the lateral compartment: extensors of the foot.
- Muscles of the posterior compartment: flexors of the toes and extensors of the foot.

II. DESCRIPTIVE ANATOMY:

A. MUSCLES OF THE ANTERIOR COMPARTMENT:

Comprising 4 muscles, from medial to lateral:

- 1. Tibialis Anterior Muscle:**
 - Origin:
 - the infracondylar tubercle of the tibia (GERDY)
 - The oblique crest of the tibial tuberosity.
 - The femoral interosseous membrane.
 - Body : long, thick, and triangular muscle
 - Insertion: on the medial cuneiform bone and the base of the first metatarsal.
 - Action: flexor, then adductor, and medial rotator of the foot.
 - Innervation: deep fibular nerve (sacral plexus).

2. **Extensor Hallucis Longus Muscle:**

- Origin: the medial surface of the fibula

The femoral interosseous membrane.

- Body: thin and flat muscle.
- Insertion: The base of the distal phalanx (P2) of the hallux.

The lateral borders of the proximal phalanx (P1) of the hallux.

- Action: Extensor of P2 on P1 and of P1 on the first metatarsal, flexor of the foot on the leg, and medial rotator of the foot.
- Innervation: Deep fibular nerve (sacral plexus).

3. **Extensor digitorum longus muscle:**

- Origin:
 - the lateral condyle of the tibia
 - The medial surface of the fibula
 - The interosseous membrane of the femur.
- Body : long and flat muscle.
- Insertion: 4 tendons supplying the last 4 toes (2nd and 3rd phalanges).
- Action: extensor of the toes, flexor and lateral rotator of the foot.
- Innervation: deep fibular nerve (sacral plexus).

4. **Peroneus tertius muscle:**

- Origin: the medial surface of the fibula

The interosseous membrane of the femur

- Body: slender and inconstant muscle.
- Insertion: the base of the 5th metatarsal.
- Action: flexor and abductor of the foot, lateral rotator of the foot.
- Innervation: deep fibular nerve (sacral plexus).

B. **MUSCLES OF THE LATERAL COMPARTMENT**

This compartment is composed of two muscles arranged in two layers:

- Deep layer: peroneus brevis
- Superficial layer: peroneus longus

1. **Peroneus brevis muscle:**

- Origin: the lateral surface of the fibula.

The anterior and posterior intermuscular septa.

- Body: short and flat muscle.
- Insertion: the lateral tuberosity of the 5th metatarsal and the 4th metatarsal.
- Action: abductor and lateral rotator of the foot.
- Innervation: superficial fibular nerve (sacral plexus).

2. **Peroneus longus muscle:**

- Origin:
 - the lateral condyle of the tibia.

- The lateral surface of the fibula.
- The anterior and posterior intermuscular septa.
- Body: thick and superficial muscle.
- Insertion: the base of the first metatarsal and the medial cuneiform bone.
- Action: extensor of the foot, abductor and lateral rotator of the foot.
- Innervation: superficial fibular nerve (sacral plexus).

C. **MUSCLES OF THE POSTERIOR COMPARTMENT:**

Comprising 8 muscles arranged in two layers

a) **Deep layer:**

1. **Popliteus muscle:**

- Origin: the lateral condyle of the femur.
- Body: short, flat, and triangular muscle.
- Insertion: Popliteal surface on the posterior aspect of the tibia, above the soleus muscle line.
- Action: Flexor and medial rotator of the leg.
- Innervation: Tibial nerve (sacral plexus).

2. **Flexor digitorum longus muscle:**

- Origin: the posterior aspect of the tibia, below the soleus muscle line (oblique line).
- Body: elongated muscle.
- Insertion: 4 tendons on the base of the distal phalanx (P3) of the last four toes.
- Action: Flexor of the toes, then extensor of the foot.
- Innervation: Tibial nerve (sacral plexus).

3. **Tibialis posterior muscle:**

- Origin:
 - Posterior surface of the tibia (upper third) below the oblique line
 - Medial surface of the fibula (upper third)
 - Posterior surface of the interosseous membrane.
- Body: Long and flat muscle.
- Insertion: Navicular bone, expansions on the three cuneiforms, the cuboid, and the base of the second, third, and fourth metatarsals.
- Action: Adductor and medial rotator of the foot.
- Innervation: Tibial nerve (sacral plexus).

4. **Flexor hallucis longus muscle:**

- Origin: Posterior surface of the fibula

Interosseous membrane.

- Body: Elongated muscle.
- Insertion: Base of the distal phalanx of the hallux (P2).
- Action: Flexes the distal phalanx (P2) on the proximal phalanx (P1), then the proximal phalanx (P1) on the first metatarsal.
- Innervation: Tibial nerve (sacral plexus).

b) Superficial layer:

Formed by the triceps surae with its three muscular heads: the soleus, the medial and lateral gastrocnemius, and the plantaris muscle.

1. Soleus muscle:

- Origin:
 - Fibular head: the head and posterior surface of the fibula.
 - Tibial head: the oblique line and the medial border of the tibia.
- Body: a large and broad muscle.
- Insertion: via the calcaneal tendon (Achilles tendon), which inserts onto the calcaneal tuberosity.
- Action: Extension of the foot, flexion of the leg at the knee, adductor and medial rotator of the foot.
- Innervation: Tibial nerve (sacral plexus).

2. Medial and lateral gastrocnemius muscles:

A large and broad muscle, consisting of two heads, medial and lateral, covering the proximal part of the soleus muscle.

- Origin: On the medial and lateral supracondylar tubercles of the femur.
- Insertion: Via the calcaneal tendon (Achilles tendon), which inserts on the calcaneal tuberosity.
- Action: Extension of the foot, flexion of the leg at the knee, adductor and medial rotator of the foot.
- Innervation: Tibial nerve (sacral plexus).

3. Plantaris muscle:

- Origin: Popliteal surface of the lateral femoral condyle.
- Body: Slender, flat, and triangular muscle.
- Insertion: Calcaneal tuberosity, medial to the Achilles tendon.
- Action: Very weak, an auxiliary muscle of the triceps surae.
- Innervation: Tibial nerve (sacral plexus).

Reference :

- Rouvière H. Anatomie Humaine Descriptive, Topographique et Fonctionnelle. Tome3.Membres.ed .Masson 2002. (traduit en anglais).
- Himmoudi SS. Le cours d'anatomie. fasc.2 appareil locomoteur. Membre inférieur. ISBN 2008. (traduit en anglais).
- Kamina .Myologie du membre inférieur.2ème édition Maloïne(traduit en anglais).

